

Documentation Cleanup Report

Date: 2026-03-15
Scope: Comprehensive documentation cleanup and gap analysis across all markdown files in docs/

1. Checkmark and Emoji Icon Removal

Problem: Checkmark emojis (✅, ❌), warning symbols (⚠️), and other Unicode icons (👉, ✖️) appeared throughout documentation, creating rendering inconsistencies across platforms.

Action: Replaced all emoji icons with plain-text equivalents:

- ✅ Text → Text (icon removed, text preserved)
- ✅ (standalone at end of line) → - PASS
- ❌ Text → No - Text
- ⚠️ Text → Note: Text
- 👉 → Pending -
- ✖️ → x

Files modified (17):

File	Icons Replaced
project_alignment_verification.md	117
phase21_assessment.md	40 + 6 warning icons
verification_log.md	40
file_inventory.md	29
research_questions_assessment.md	13
eda_and_data_split_summary.md	12
cbd_comparison_and_validity_report.md	10
service_model_spec.md	10
final_summary.md	9
gap_closure_report.md	9
gap_remediation_plan.md	8 + 3 warning icons
technical_report.md	8
phase_comparison_summary.md	7
optimization_results.md	6 + 2 cross marks
executive_presentation.md	4
source_manifest.md	3
firehouse_capacity_analysis.md	2 + 3 warning icons + 1 pending icon

Total icons replaced: ~327

2. K=30 Capacity Analysis Added to Section 5.12

Problem: Section 5.12 of `technical_report.md` documented capacity sensitivity only for K=20 and K=40, despite K=30 simulation data existing in `results/capacity_comparison/simulation_results_K30.csv`.

Action:

- Added new subsection “K = 30: Capacity Remains Largely Non-Binding” between K=20 and K=40
- Included performance table with Cap=1/2/3/5 results for all three policies
- Updated intro paragraph: “2 K values” → “3 K values” (K=20, K=30, K=40)
- Updated experiment count: “30 experiments” → “45 experiments”
- Updated Figure 6 caption to reference K=30

Key K=30 findings documented:

- P0-spatial: 2.78 min mean RT, constant across all capacity levels
- P1: 2.47 min (cap=1) to 2.39 min (cap=2), < 0.09 min variation
- P2: 2.43–2.50 min range, < 0.08 min variation
- Confirms capacity constraints are non-binding for fleets ≤ 30

3. Image Link Audit

Total image references scanned: 24 across 5 markdown files

Results:

- All 24 image links use correct relative paths (e.g., `../results/figures/...`)
- All 24 referenced image files exist on disk
- No absolute paths (`/home/ubuntu/...`) found in image references
- One false positive: an inline code example `![CBD Scenario Comparison](/home/ubuntu/ems-optimization/docs/...)` in `phase21_assessment.md` (not an actual image reference)

Files with image references:

File	Image Count	Status
technical_report.md	12	All valid
capacity_sensitivity_analysis.md	4	All valid
queue_analysis.md	4	All valid
cbd_robustness_analysis.md	3	All valid
phase21_assessment.md	1 (example only)	N/A

4. Comprehensive Documentation Review

Internal document links: All verified — no broken cross-references found.

Key content completeness checks:

Content Area	Status
All three policies (P0, P1, P2) documented	Complete (73, 29, 75 mentions)
CBD analysis	Complete (78 mentions)

Capacity analysis (K=20, K=30, K=40)	Complete (all three K values now documented)
Fleet sizes K=15 through K=45	Complete (K=15: 7, K=20: 26, K=30: 6, K=40: 12 mentions)
All referenced figures exist	Verified

Additional cleanups:

- Removed stray Unicode cross marks (✖) from optimization_results.md
- Replaced pending emoji (👉) in firehouse_capacity_analysis.md
- No absolute file paths found in documentation text
- No broken internal document links detected

Summary

Category	Issues Found	Issues Fixed
Checkmark/emoji icons	327+ across 17 files	All replaced with plain text
K=30 missing from §5.12	1 section gap	Added with data table
Broken image links	0	N/A
Other emoji/Unicode	12 (⚠️, 👉, ✖)	All replaced
Broken internal links	0	N/A
Incomplete sections	0	N/A

All documentation is now clean, complete, and uses consistent plain-text formatting.